We conduct a series of experiments in which we execute our forecaster on the cross product of: with and without search, and with and without market prices on ForecastBench. We present these results in Table 4.

We find that simply providing the forecaster with prediction market prices improves the performance of a no-news baseline from 0.116 to 0.103 on the subset of ForecastBench derived from public prediction markets—an 11.2% improvement. Providing the market price to our best-performing configuration improves performance from 0.085 to 0.075 on the same subset, a similar 11.8% improvement. In these experiments, we prevent the AIA Forecaster from retrieving prediction market prices unless they are explicitly provided.[11]

Includes Market Price?	Forecasting Method	Brier score
×	LLM, no search	0.116
✓	LLM, no search	0.103
✓	Market price only	0.096
×	LLM, agentic search	0.085
✓	LLM, agentic search	**0.075**

Table 4: The interaction of market prices and search. Providing the LLM with access to market prices closes $\sim 42\%$ of the gap between the no-search and agentic search baselines. This suggests that the impact of search is strongly mitigated by access to market prices, which effectively summarize large volumes of relevant information.

We contrast these to the results of Karger et al. (2024), who find that the best-performing LLM-based forecaster also requires access to market prices and achieves a Brier score of 0.109 on this subset.[12] However, the market price alone achieves a Brier score of 0.096 on the same subset of questions; that is, the best-performing LLM-based forecaster in Karger et al. (2024) would perform better if it simply *output the market price which is provided in its prompt*. The AIA Forecaster also benefits from access to the market price, but performs significantly better than using the market price alone (0.096 for the market price versus 0.075 for the AIA Forecaster).

Taken together, these results suggest caution in interpreting the performance of LLM-based forecasting systems which have direct access to prediction market prices. On the one hand, these markets can be highly efficient information aggregators, and human and artificial forecasters can both benefit from incorporating these prices into their own predictions. Prediction markets also serve as a convenient testbed for evaluating forecasting performance. On the other hand, access to market prices can mask important differences in forecasting capability, particularly when extrapolating performance to domains where no such prices are available.

11. Specifically, we blacklist domains associated with the largest public prediction markets.
12. Karger et al. (2024) report that the best overall forecaster achieves a Brier score of 0.109 on this subset. The model which performs best on this subset of questions (but worse on the full benchmark) achieves a score of 0.096 on this subset.

5.2 Mitigating Foreknowledge Bias in Search

We now turn to the problem of foreknowledge bias. As discussed above, evaluating search-augmented forecasting systems presents a fundamental challenge: search providers can inadvertently leak information about events that have already occurred. In the extreme case, this bias can create the illusion of forecasting skill where none exists.

For example, consider evaluating an LLM forecaster on the following event: "Will Alphabet's stock price reach $190 by December 2024?" The resolution for this event is already known as of this writing (Alphabet stock crossed the $190 threshold in December 2024), which allows us to evaluate forecasting skill by, e.g., blinding the LLM to events occurring after July 2024 and asking it to provide a forecast. Unfortunately, if the forecaster generates a benign-sounding search query like "Google stock price analysis", the search provider may return an article simply stating the factual outcome of this event. This can occur even if the search provider is instructed not to return any articles published after July 2024 — as it turns out, determining the date when an article was published (and, perhaps, subsequently updated) is a challenging problem.

In this section, we estimate the prevalence of foreknowledge bias and bound its effect on our results. Through manual analysis, we taxonomize sources of foreknowledge bias and provide examples in Appendix C. We then develop an LLM-as-a-judge pipeline that reviews search provider responses and flags those that exhibit foreknowledge bias.

For each query, we collect search provider snippets and any passages cited by the forecaster, prompt the judge to indicate whether the evidence implies knowledge beyond the intended date cutoff, and record a binary flag with a brief rationale. The exact prompt and JSON schema appear in Appendix D.

Validation of the judge. We manually audit n=502 search provider traces randomly sampled across both flagged and unflagged items. Table 5 summarizes performance. As intended for a high-recall detector, flags are noisy positives (low precision), while "clean" classifications are highly reliable (low false negative rate). Practically, this makes it relatively safe to trust unflagged runs, while treating raw flag counts as an overestimate of true foreknowledge bias.

Classification (n=502)	**Count**	**Percentage**
True Positives (correctly identified leaks)	16	3.2%
False Positives (flagged clean responses)	65	12.9%
True Negatives (correctly identified clean)	412	82.1%
False Negatives (missed leaks)	9	1.8%

Table 5: Validation of the judge on 502 randomly sampled traces. While the flags are noisy (a relatively high false positive rate), responses which are not flagged reliably exclude foreknowledge (low false negative rate).

From this exercise, we estimate that $\sim 1.65\%$ of all search results include some form of foreknowledge bias. We provide additional details for this calculation in Appendix E. With this estimate in hand, we now seek to bound the impact of foreknowledge bias on forecasting performance.

Robustness checks. We test whether residual foreknowledge bias affects our performance metrics through two conservative robustness checks:

- **Filtered (all flags):** Remove every judge-flagged response. Roughly 80% of removed forecasts do not exhibit foreknowledge bias, making this test intentionally conservative.
- **Worst-case:** Assume a forecast of 0.50 (corresponding to a Brier score of 0.25) for any question with $\geq$5 flags. This is again intentionally conservative, assuming no forecasting skill on any questions which appear to suffer from foreknowledge bias.

We report performance without the consistency analysis and statistical correction steps to isolate the effect of foreknowledge bias.

Analysis	Brier score	Change
Baseline	0.1159	—
Filtered	0.1161	+0.17%
Worst-case	0.1166	+0.55%

Table 6: Robustness checks to evaluate foreknowledge bias on ForecastBench. "Filtered" computes the Brier score after removing all questions flagged by the judge. "Worst-case" assumes a forecast of 0.5 for all flagged questions. Both perform comparably to a no-mitigation baseline.

As Table 6 demonstrates, both robustness checks yield a Brier score that is within 0.001 of the baseline value ($\leq 0.6\%$ on a relative basis). Thus, we conclude that foreknowledge bias does not appear to materially affect performance.

5.3 LLM Forecasts Are Diversifying to Market Prices

We now investigate the complementarity between LLM forecasts and market prices. In Section 4 we saw that our forecaster outperforms the market consensus on ForecastBench, but modestly underperforms on the stronger MarketLiquid benchmark. It is natural to then ask: in cases where the forecaster outperforms the market, does the market price provide any *additional* predictive signal? And conversely, in settings where the forecaster underperforms the market consensus, can we disregard its forecasts entirely?

We investigate these questions by estimating a bivariate simplex-constrained regression of the binary resolution indicator on two covariates: the market price and the LLM forecast. This regression learns a convex combination of the two covariates that minimizes the